GenCC Recommendation for Dominant Negative (DN) mechanism curation - Version 1.0

The Gene Curation Coalition proposes a semi-quantitative metric to determine the mechanism of disease for gene-disease relationships. The framework is intended to be used for human gene-disease relationships that have scored at Moderate or higher using the ClinGen gene curation framework (Strande et al 2017, see here for the most recent gene curation SOP). Levels should be capped at "Suspected" if applying to genes with Limited or below evidence. Caution should be exercised if adapting to non-human genetic diseases as this framework has not been tested using non-human phenotypes.

This framework is for evaluating genes for a dominant negative mechanism.

**While this is a semi-quantitative framework, please note that points are not statistical in nature, nor do they correspond to Bayesian points in the sequence variant classification frameworks.

Table 1. Scoring Matrix for Dominant Negative Framework
For this framework, the inheritance MUST be autosomal dominant or de novo

Variant Type[1]	**Evidence**	**Points**
DN Variant[2]	Strong functional variant-level evidence demonstrating a DN effect	0.15 pt (range 0.1-0.2)
Experimental Evidence	KI or transgenic model organism with consistent phenotype and zygosity[3]	0.3 pt maximum cap (range 0.1-0.3)
	Known function of the gene supports a DN mechanism[4]	0.1-0.3pts
	Highly homologous protein has a well-established DN mechanism[5]	0.1-0.3 pts, depending on the evidence and number of proteins
	Human patients with KO variants have a directionally concordant phenotype OR (if no human data are available) KO animal model has a directionally concordant phenotype	0.2 pts OR 0.1 pts

Variant spectrum	Presence of >3 recurrent variants in a cluster (somatic included)	0.1 pts

Footnotes:
[1]Variants can only be counted when part of a case that has variants of the types above that are assumed to be causal based on consistent phenotype. Each variant can only be scored once; even if that variant has been observed in recurrent cases.
[2]DN variants are usually (but may not be limited to) missense variants, variants that cause in-frame deletions or insertions, or truncating variants that are predicted to escape NMD. In this case the DN allele MUST be expressed to be considered for DN.
[3]This animal model MUST demonstrate a DN mechanism. If it does not specifically demonstrate DN, it can be used for other experiments that demonstrate this mechanism (e.g. ex vivo experiments on tissues or cells from the organism).
[4]Examples of this would be a gene that is part of a functional complex that forms a multimer
[5]This scoring applies to both orthologs and paralogs. Some well-known examples are Collagen genes and Gap Junction protein genes.

Abbreviations: DN, Dominant Negative; KI, Knock in; KO, knock out; NMD, nonsense-mediated decay

Table 2. Summary Matrix for Dominant Negative Framework

Points from Table 1 should be summed and the equivalent confidence term applied to the DN mechanism.

Evidence Points	Confidence Term	Mechanism
.99 or higher	Established	Dominant Negative
0.90-0.98	Likely	
0.5-0.89	Suspected	
0.49-0	Uncertain	